
POWER IN — USB-C PD sink (15 V)
CH224K DP/DM shorted = PD-only; Type-C D± go to the MCU.
CFG1=56k -> 15 V PDO; falls back to 5 V (charger idles <11.2 V).
USB 2.0 subset of the 24-pin USB4160 (SS pads unconnected).
D+/D- = native USB-JTAG (IO19/20). Vertical - port out the back wall.
CHARGER — LT3652 (1S buck, 4.05 V float, 1.0 A)
Float 4.05 V (R14/R15): FULLCHG_EN -> 4.20 V full-charge mode
I_CHG = 1.0 A (R18): RT1 = NCP18XH103 on the holder, 0.45 C window;
Charges only on the 15 V contract (UVLO 11.2 V): runs with no cell.
MCU — ESP32-S3-WROOM-1-N16R8 (Wi-Fi + BLE, native USB-JTAG)
IO35-37 reserved by the octal PSRAM (N16R8). Straps: IO0 high; IO45/46 low via
their wake gate nets + internal PD; IO3 internal. Flash+console+JTAG = USB-CDC (IO19/20).
STORAGE — microSD (DM3AT), SPI2 dedicated
Layout: C222/C223 at the J6 VDD pin.
SPI mode: MSB-first, CS active-low, ~25 MHz. No card-detect (no spare GPIO).
BATTERY — 18650 + reverse P-FET + protector + TCO
Li-Ion 18650 ONLY - 2.5-4.2 V
18650 (Keystone 1043)
cell- -> AOSD32334C (DD/OC gated) -> TCO 77 C -> PACK- (GND)
OV 4.28 V / OD 2.90 V fixed (-G8). Independent of the charger.
RAILS — 5 V boost (always on) -> 3V3 buck
+5V: display panel, stepper VM, panel LEDs, PVDD-mux aux. +3V3: MCU + all logic.
SENSORS (I2C off-board) + HAND-HOMING (QRE1113)
Sensor board (BME688+TSL2591+LIS3DH)
QRE1113GR faces reflective tabs on the hand hubs (no magnets).
Sensor breakouts carry their own I2C pull-ups.
AUDIO — TAS5760M PBTL + LC filter + PVDD mux (12V/5V)
PVDD auto-mux: 12 V wins when plugged; on battery Q4 ideal-diodes
the 5 V rail -> ~3 W quieter alarm. Firmware: LEDs+audio <= ~12 W.
PBTL mono (reg 0x06[7]=1): OUTA||OUTB. MCLK 256fs ~12.288 MHz.
HOW TO READ THIS SHEET
One page, laid out like block_diagram.drawio: power flows left,
MCU centre, peripherals right. Power paths and MCU buses (SPI,
I2S, MCPWM, encoder) are drawn as wires; GND / +3V3 / +5V taps
and named labels connect by name (same name = same net).
Off-board parts (SD card, sensors, speaker, wake-LED strips,
knob, radio toggle, cell) enter via connectors J1..J11 / BT1.
RAIL — 12 V boost (plugged-only: audio PVDD + wake LEDs)
~12 W ceiling from 15 input: wake LEDs + audio share it.
PLUGGED-ONLY: firmware asserts BOOST12_EN only while PD_PG is live.
LED — wake COB drivers (12 V, plugged-only)
Wake COB 12V (warm+cool)
Wake COB strips are self-ballasted (12 V, plugged-only): firmware
gates their PWM off on battery. ~1 kHz, gamma; LED + audio <= ~12 W.
DIAL NEOPIXEL HEAD — 2x on-PCB + 5x off-board via J12 (5 V)
D40/D41 = on-PCB dial wash (chain pos 1-2). J12 breaks the chain out
for 5 more pixels wired off-board (status row, chain pos 3-7). One RMT
data line; AHCT buffer lifts 3V3 -> 5V (SK6812 VIH = 0.7*VDD = 3.5V).
NeoPixel ext (5V/DATA/GND, pos 3-7)
Layout: D40/D41 = dial wash either side of the shaft. C230/231 one at
each pixel VDD; C237 bulk + R110 + U15 at the chain head. J12 feeds
5 more SK6812 wired off-board in series (chain pos 3-7), same rail.
IO EXPANDER (MCP23017 @0x20) + KNOB (EM14)
Radio-off toggle (rear)
Knob: EM14 enc+SW (top face)
INT: any GPA/GPB change -> IO44 (IOCON.MIRROR=1). Knob = EM14A0D-C24-L064S
(optical, 64 CPR, no detent, push) on the top face. J7/J10 = JST ZH 1x06, J11 = 2H
1x02 (pre-crimped AxxZR cables). A/B 5 V -> 100k/200k div -> PCNT; SW -> IO17 IRQ.
MOTOR — 2x TB6612FNG -> X40.879 dual-shaft stepper
Layout: each driver's 10uF+100nF go at its VM pins, the 100nF at its VCC pin.
PWM-on-IN microstepping: MCPWM waveforms on AIN/BIN, PWMA/B tied high.
STBY_STBY (expander GPA1) low at boot -> drivers off. Call map: esp32.md.
Tibo Mauron
Sheet: /
File: clock.kicad_sch
Title: Wooden Clock — main board
Size: A1 Date: 2026-07-14 Rev: 0.2
KiCad E.D.A. 10.0.4 Id: 1/1